WORKED SOLUTION

STEP II 2006, Question 4

Pure • Integration by Substitution

STEP II 2006, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q4

The key idea

The substitution $x = \pi - t$ leaves $\sin x$ unchanged but turns the factor x into $\pi - t$, so the integral reappears on the right-hand side and can be solved for, like an equation. This removes the awkward factor x entirely. After that, every integral in the question is a function of $\sin x$ times $\sin x$ (or $\cos x$), and a substitution $u = \cos x$ or $u = \sin x$ finishes the job. The real work is noticing how to bring parts (ii) and (iii) into the form the identity covers.

Solution

The identity

Let $I = \int_0^\pi x\, f(\sin x)\, \mathrm{d}x$. Put $x = \pi - t$, so $\mathrm{d}x = -\mathrm{d}t$; when $x = 0$, $t = \pi$, and when $x = \pi$, $t = 0$. Since $\sin(\pi - t) = \sin t$,

$$\begin{aligned} I &= \int_\pi^0 (\pi - t)\, f(\sin t)\, (-\mathrm{d}t) = \int_0^\pi (\pi - t)\, f(\sin t)\, \mathrm{d}t \\ &= \pi \int_0^\pi f(\sin t)\, \mathrm{d}t - \int_0^\pi t\, f(\sin t)\, \mathrm{d}t = \pi \int_0^\pi f(\sin x)\, \mathrm{d}x - I, \end{aligned}$$

because the name of the integration variable doesn't matter. Hence $2I = \pi \int_0^\pi f(\sin x)\, \mathrm{d}x$, which is the result we needed:

$$\int_0^\pi x\, f(\sin x)\, \mathrm{d}x = \tfrac{1}{2}\pi \int_0^\pi f(\sin x)\, \mathrm{d}x.$$

Part (i)

Here $f(\sin x) = \dfrac{\sin x}{3 + \sin^2 x}$, so the identity gives

$$\int_0^\pi \frac{x \sin x}{3 + \sin^2 x}\, \mathrm{d}x = \tfrac{1}{2}\pi J, \qquad J = \int_0^\pi \frac{\sin x}{3 + \sin^2 x}\, \mathrm{d}x.$$

Write $3+\sin^2 x = 4-\cos^2 x$ and substitute $u=\cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$, with u running from 1 down to -1:

$$J = \int_{-1}^{1} \frac{\mathrm{d}u}{4-u^2}.$$

By partial fractions, $\dfrac{1}{4-u^2} = \dfrac{1}{4}\left(\dfrac{1}{2-u}+\dfrac{1}{2+u}\right)$, so

$$J = \frac{1}{4}\left[\ln\frac{2+u}{2-u}\right]_{-1}^{1} = \frac{1}{4}(\ln 3 - \ln \tfrac{1}{3}) = \tfrac{1}{2}\ln 3.$$

(Both $2\pm u$ are positive on $[-1,1]$, so no modulus signs are needed.) Therefore

$$\int_0^{\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x = \frac{\pi}{4}\ln 3.$$

Part (ii)

Ask yourself The identity only covers $[0,\pi]$. How do we deal with $[\pi,2\pi]$?

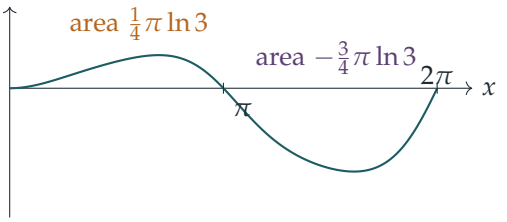
Split the range at π and shift the second piece back to $[0,\pi]$ with $x=\pi+t$. Then $\sin x = -\sin t$ and $\sin^2 x = \sin^2 t$, so

$$\begin{aligned}\int_{\pi}^{2\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x &= -\int_0^{\pi} \frac{(\pi+t)\sin t}{3+\sin^2 t}\,\mathrm{d}t = -\pi J - \int_0^{\pi} \frac{t\sin t}{3+\sin^2 t}\,\mathrm{d}t \\ &= -\tfrac{1}{2}\pi\ln 3 - \tfrac{1}{4}\pi\ln 3 = -\tfrac{3}{4}\pi\ln 3,\end{aligned}$$

using $J=\frac{1}{2}\ln 3$ and part (i). Adding the integral over $[0,\pi]$ from part (i),

$$\int_0^{2\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x = \frac{\pi}{4}\ln 3 - \frac{3\pi}{4}\ln 3 = -\frac{\pi}{2}\ln 3.$$

The sketch shows why the answer is negative: the integrand has the same shape on each half, but on $[\pi,2\pi]$ it is weighted by larger values of x.



Part (iii)

Since $\sin 2x = 2\sin x\cos x$ and $\sin x \geq 0$ on $[0,\pi]$,

$$|\sin 2x| = 2\sin x\,|\cos x| = 2\sin x\sqrt{1-\sin^2 x},$$

which is a function of $\sin x$ alone.

Ask yourself Why is it important that $|\cos x|$, not $\cos x$, appears here?

On $[0, \pi]$, $\cos x$ is *not* a function of $\sin x$: $\cos(\pi - x) = -\cos x$ while $\sin(\pi - x) = \sin x$. But $|\cos x| = \sqrt{1 - \sin^2 x}$ is, so the modulus is exactly what lets us use the identity:

$$\int_0^{\pi} \frac{x|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = \tfrac{1}{2}\pi \int_0^{\pi} \frac{|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x.$$

The new integrand is unchanged by $x \mapsto \pi - x$, so the integral over $[0, \pi]$ is twice that over $[0, \frac{1}{2}\pi]$, where $\cos x \geq 0$ and the modulus can be dropped:

$$\int_0^{\pi} \frac{|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = 2\int_0^{\pi/2} \frac{2\sin x \cos x}{3 + \sin^2 x}\,\mathrm{d}x = 2\Big[\ln(3 + \sin^2 x)\Big]_0^{\pi/2} = 2\ln\tfrac{4}{3},$$

since the numerator is the derivative of the denominator. Therefore

$$\int_0^{\pi} \frac{x|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = \tfrac{1}{2}\pi \cdot 2\ln\tfrac{4}{3} = \pi \ln\frac{4}{3}.$$

Checks

- Simpson's rule gives 0.86285 for (i), -1.72570 for (ii) and 0.90378 for (iii), matching $\frac{1}{4}\pi\ln 3$, $-\frac{1}{2}\pi\ln 3$ and $\pi\ln\frac{4}{3}$.
- Answer (ii) is exactly -2 times answer (i). Splitting at π explains this: the two halves are $\frac{1}{4}\pi\ln 3$ and $-\frac{3}{4}\pi\ln 3$.
- A quick test of the identity: with $f \equiv 1$ both sides equal $\frac{1}{2}\pi^2$.

Common slips

- Using the analogous "identity" on $[0, 2\pi]$, which would give $\pi\int_0^{2\pi} f(\sin x)\,\mathrm{d}x = 0$. The proof needs $\sin(2\pi - t) = \sin t$, which is false.
- In (iii), writing $|\sin 2x| = 2\sin x\cos x$ throughout $[0, \pi]$, which gives 0 for the inner integral.

Beyond STEP

The trick of making an integral reappear and solving for it, and the use of symmetries like $x \mapsto \pi - x$, are standard tools in analysis and in evaluating definite integrals that have no elementary antiderivative. They lead to Fourier series, where integrals of x times trigonometric functions over a period are the basic objects. For more on this style of argument, see Siklos, *Advanced Problems in Mathematics*, and for the rigorous theory of the integral, Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.